

Jiaqi Zhang

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EDUCATION

McGill University | Montreal, QC

B.Sc. in Mathematics and Computer Science

Expected May 2028

Relevant Coursework: Linear Algebra, Probability, Abstract Algebra, Algorithms, Real Analysis

RESEARCH INTERESTS

Cryptographic protocols; threshold cryptography; secure computation; algorithmic game theory; rational and adversarial behavior

RESEARCH EXPERIENCE

Concordia University | Montreal, QC

Research Assistant

Jan 2026 - Present

- Contribute to research in security and cryptographic protocols through protocol analysis, implementation, and experimental evaluation.

Adrian Vetta Research Group, McGill University

Research Group Participant

Current

- Participate in research meetings and discussions within Prof. Adrian Vetta's group, with emphasis on formal reasoning about incentives and strategic behavior.

SELECTED RESEARCH

Verify-or-Scar: Bribery Resistance for Weighted Threshold Services

Manuscript under review

Independently completed the research and manuscript, with guidance from Prof. Honghao Fu.

- Studied bribery in weighted threshold services where an attacker can verify a disclosed contribution before deciding whether to purchase it.
- Designed a mechanism in which trusted external verification triggers persistent liability, and analyzed procurement and participation incentives under rational behavior.
- Implemented and evaluated the mechanism in a trusted-execution setting, testing sanction persistence, retries, recovery, non-purchase, and threshold composition.

SELECTED PROJECT

Road Sign Classification with Denoising Pipeline

- Developed a noise-aware classification pipeline with multi-stage denoising and feature enhancement to study the effect of preprocessing on model performance.
- Analyzed noise characteristics and misclassification patterns to guide preprocessing and algorithmic improvements.

TECHNICAL SKILLS

Programming & Data: Python, Java, NumPy, JSON

Security & Systems: Cryptographic protocol analysis, threshold systems, trusted execution environments (TEEs), remote attestation, Azure TDX

Research & Tooling: Mathematical and security modeling, algorithmic analysis, protocol implementation, experimental design, failure and recovery testing, Git, GitHub